

Compensated Gain-Layer Design and Defect-Aware Modeling of Irradiated 4H-SiC LGADs

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Abstract

4H-SiC Low-Gain Avalanche Detectors (LGADs) have emerged as highly promising candidates for radiation-hard timing detectors due to their wide bandgap, high critical electric field, high carrier saturation velocity, and exceptionally low leakage current under high-bias operation. These intrinsic properties make them attractive for future collider upgrades, such as the High-Luminosity Large Hadron Collider (HL-LHC), where radiation fluences can exceed $10^{15} \text{ n}_{\text{eq}}/\text{cm}^2$. In this work, we present a Silvaco Victory TCAD investigation of thin and ultra-thin 4H-SiC LGAD structures optimized for precision timing in high-radiation environments. Different impact ionization models, including the Hatakeyama, Konstantinov, and Loh formulations, are systematically benchmarked against recently reported experimental I - V , C - V , and gain measurements from SICAR and BNL/OnSemi devices to rigorously evaluate their suitability for avalanche multiplication modeling in 4H-SiC. Radiation-induced degradation is studied through the implementation of deep-level defect states commonly observed in irradiated SiC, and their subsequent effects on electric field redistribution, gain suppression, leakage current, and charge collection efficiency (CCE) are systematically investigated for both proton and neutron irradiation conditions. To improve high-field stability, an etched Junction Termination Extension (JTE) structure is incorporated to suppress premature edge breakdown while sustaining peak electric fields approaching $\sim 3 \text{ MV/cm}$. In parallel, a compensated gain-layer design based on donor-acceptor co-doping is introduced to reduce effective gain-layer degradation after irradiation and improve multiplication stability. Preliminary simulation results indicate improved preservation of effective doping, enhanced post-irradiation gain stability, and an extended operational bias range compared to conventional LGAD structures. Finally, time resolution studies are performed to evaluate the timing performance of the proposed 4H-SiC designs in comparison with conventional silicon LGAD detectors.

Keywords: 4H-SiC LGAD, Radiation Damage, TCAD Simulation, Avalanche Multiplication, Precision Timing Detectors

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